JOSEF DOORNINK

Site Reliability Engineer | AI Infrastructure & MLOps

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PROFESSIONAL SUMMARY

A creative problem-solver who loves building the infrastructure that brings ideas into reality. An engineer with 10+ years experience, 10 peer-reviewed papers, 2 US patents, a major design award and FDA-cleared hardware. That foundation now drives deep SRE and AI infrastructure work (CKS/CKA certified), delivering systems with integrity and observability at scale - currently designing autonomous Agentic SRE pipelines that leverage LLMs for root-cause analysis.

TECHNICAL SKILLS

Core Technologies: Kubernetes (AKS) | Docker | Terraform | Azure | AWS | GCP | Helm | YAML | Python | Go/Golang | Bash | C# | SQL | Kafka | Redis

GPU & LLM Serving: vLLM | GPU Sizing & Capacity Planning | KV Cache Tuning | Model Quantization (AWQ) | KEDA / HPA Autoscaling | GPU Node Pools (AKS)

AI/ML Engineering: LLM Model Serving | ML Pipelines | Distributed Training Systems | Azure Machine Learning | Vertex AI | Drift Detection | Model Finetuning Systems

Observability & SRE: New Relic | Azure Monitor | Distributed Tracing | CI/CD Pipelines | GitHub Actions | Performance Profiling | Incident Response | Prometheus | Grafana | Azure DevOps

PROFESSIONAL EXPERIENCE - STARTUP

Lead MLOps Engineer | Reason Benefit AI Corporation | Remote | Oct 2025 - Feb 2026

- Architected and maintained large-scale AKS production environments for ML model training, supporting distributed model inference at scale
- Improved inference throughput by 40% by optimizing model serving infrastructure through systematic performance profiling and tuning
- Reduced training cycle times by profiling RL pipeline bottlenecks by introducing automated pipeline scripts and validations

PROFESSIONAL EXPERIENCE

Lead Site Reliability Engineer (SRE) I -> II -> III | Trimble | Portland, OR | Jan 2019 - Present

- Achieved 99.9% uptime SLA managing 10M+ requests/day by architecting and scaling AKS production environments across 30+ distributed microservices
- Eliminated 80+ hours/month of operational toil and accelerated deployment velocity 3x by developing high-performance automation tools in Python and Go
- Accelerated model deployment velocity and reduced manual overhead by 70% by building yaml-based automation tools for ML pipeline orchestration
- Reduced P99 latency by 45% and improved throughput by 60% by leading systematic performance optimization and instrumentation initiatives
- Streamlined workflows and improved developer productivity for 50+ engineers by building custom CI/CD pipelines and Go CLI tooling (Cobra)
- Reduced MTTR by 50% by designing and implementing a comprehensive observability stack with distributed tracing in New Relic
- Managed 500+ cloud resources consistently by implementing Infrastructure as Code methodologies using Terraform

Software Developer | Viewpoint | Portland, OR | Mar 2018 - Jan 2019

- Scaled multi-tenant architectures to serve thousands of users by developing robust RESTful APIs and migrating on-premise SaaS solutions to Azure (.NET/Angular)

Software Developer I | Onfulfillment | Portland, OR | Mar 2014 - Mar 2018

- Improved application response times by 40% (measured via New Relic APM) by engineering and leading the uplift migration of a legacy .NET/C# monolith to a modern greenfield platform

Biomechanical Research Engineer II | Legacy Biomechanics Research Lab | Portland, OR | Jan 2007 - Dec 2013

- Delivered multiple US FDA-approved internal implants by leading the test methodology and development lifecycle for a multimillion-dollar NIH-funded orthopaedic project

KEY TECHNICAL PROJECTS

vLLM on Kubernetes: GPU Sizing & Autoscaling (2026) - Served a quantized Qwen2.5-7B model on vLLM across A10 GPU nodes in AKS. Derived GPU sizing from first principles (weight memory vs. KV cache headroom) and eliminated autoscaler flapping by scaling KEDA on a leading KV-cache-utilization signal instead of lagging queue depth. Published as an article series on Dev.to.

K8gentS (Kubernetes RCA Agent) (2026) - Created an autonomous Root Cause Analysis tool bridging SRE and AI. Leverages LLM reasoning and system telemetry to automatically diagnose pod failures, drastically reducing MTTR footprint.

OmniSight-Core (2026) - Engineered a centralized observability platform and video search engine capable of understanding semantic queries, utilizing self-healing infrastructure to automatically detect model performance decay.

Distributed Training Pipeline Optimization (2025) - Improved training cycle time by profiling RL pipeline bottlenecks, reducing Python GIL contention, and introducing automated pipeline validation.

Kubernetes Security Hardening (2024) - Reduced security risk by implementing CKS controls and automation that supported SOC2 compliance.

High-Performance CLI Tooling (2023) - Created production-grade CLI tools in Go for infrastructure management adopted by 50+ engineers.

CERTS/COURSES

- **CNCF Certified Kubernetes Security Specialist (CKS)** | March 2024 | LF-ghgugl1a0s
- **CNCF Certified Kubernetes Administrator (CKA)** | June 2021 | LF-w50bpv1lpd
- **Machine Learning Specialization** | Stanford/Coursera | September 2025
- **Microsoft Certified Azure Developer Associate** | August 2019 | H210-5692
- **HashiCorp Certified Terraform Associate** | July 2022 | HCTAO-002
- **Production Machine Learning Systems** | Google Cloud / Coursera | April 2026 | 01X41BH2KFVE

EDUCATION

Master of Science, University of California, Davis | 2006

Bachelor of Science, Mechanical Engineering | California State University, Chico | 2003

PUBLICATIONS

- M Bottlang, **J Doornink**, DC Fitzpatrick, SM Madey. *Far cortical locking can reduce stiffness of locked plating constructs while retaining construct strength.* The Journal of Bone and Joint Surgery (JBJS), 2009
- M Bottlang, **J Doornink**, GD Byrd, DC Fitzpatrick, SM Madey. *A nonlocking end screw can decrease fracture risk caused by locked plating in the osteoporotic diaphysis.* The Journal of Bone and Joint Surgery (JBJS), 2009
- DC Fitzpatrick, **J Doornink**, SM Madey, M Bottlang. *Relative stability of conventional and locked plating fixation in a model of the osteoporotic femoral diaphysis.* Clinical Biomechanics, 2009
- M Bottlang, M Lesser, J Koerber, **J Doornink**, B von Rechenberg, P Augat, DC Fitzpatrick, SM Madey, JL Marsh. *Far cortical locking can improve healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- M Bottlang, **J Doornink**, TJ Lujan, DC Fitzpatrick, JL Marsh, P Augat, B von Rechenberg, M Lesser, SM Madey. *Effects of construct stiffness on healing of fractures stabilized with locking plates.* The Journal of Bone and Joint Surgery (JBJS), 2010
- **J Doornink**, DC Fitzpatrick, S Boldhaus, SM Madey, M Bottlang. *Effects of hybrid plating with locked and nonlocked screws on the strength of locked plating constructs in the osteoporotic diaphysis.* Journal of Trauma and Acute Care Surgery, 2010
- **J Doornink**, DC Fitzpatrick, SM Madey, M Bottlang. *Far cortical locking enables flexible fixation with periarticular locking plates.* Journal of Orthopaedic Trauma, 2011
- PJ Denard, **J Doornink**, D Phelan, SM Madey, DC Fitzpatrick, M Bottlang. *Biplanar fixation of a locking plate in the diaphysis improves construct strength.* Clinical Biomechanics, 2011
- PA Wackym, JA Ratigan, JD Birck, SH Johnson, **J Doornink**, M Bottlang, SK Gardiner, FO Black. *Rapid cVEMP and oVEMP responses elicited by a novel head striker and recording device.* Otology & Neurotology, 2012
- M Bottlang, DC Fitzpatrick, D Sheerin, E Kubiak, R Gellman, C Vande Zandschulp, **J Doornink**, K Earley, SM Madey. *Dynamic fixation of distal femur fractures using far cortical locking screws: a prospective observational study.* Journal of Orthopaedic Trauma, 2014

PATENTS & AWARDS

Patents:

- *Bone Screw with Multiple Thread Profiles for Far Cortical Locking and Flexible Engagement to a Bone.* US Patent No. US10918430B2 (2021), US9314286B2 (2016), US8740955B2 (2014)

Awards:

- *Scientific Exhibit Award of Excellence*, American Academy of Orthopaedic Surgeons (AAOS), 2010

FDA-Approved Medical Devices:

- Led engineering team that created 4 FDA-approved orthopaedic implants currently used in national and international trauma centers (K101696, K123918, K130810)